

## Flat Earth Celestial Navigation Model

The model I will be using here is similar to Gleason's map. In both cases, latitude converts to a distance from the center (North Pole), and longitudes are polar coordinates. However, my model uses an infinite plane, and unlike Gleason's map, it does not have a linear relationship between distance from the pole and latitude. Instead, the correct way to describe distance from the pole in my model is a formula I discovered through trial and error:

$$d(\varphi) = \frac{\cos(\varphi)}{1 + \sin(\varphi)}$$

The exact output unit is unknown, since I do not know the true sizes of cities and countries. However, this does not really matter, since the unit will be used consistently throughout the whole method. Changing this unit would only scale the map of Earth as a whole; it would not change the results.

The next important formula is the one that converts latitude and longitude into Cartesian coordinates  $(x, y)$ . Since longitude is a polar coordinate, the North Pole will sit at the center of the coordinate system. I use the  $x$ -axis as the Greenwich meridian. From these definitions the following equations hold:

$$x = \cos(\lambda) \cdot d(\varphi)$$

$$y = \sin(\lambda) \cdot d(\varphi)$$

Now that we know how latitude and longitude translate to our flat coordinate system, we can work exclusively with planar math.

### Geographical Position (GP) of a Star

To work out the geographical position (GP) of a star, we have to understand the meaning of some terms in the nautical almanac:

- **Declination (Dec):** Latitude of the GP of the star.
- **GHA (Greenwich Hour Angle):** Same as longitude, but measured westwards.
- **Aries GHA:** GHA of the first point of Aries, an arbitrary celestial reference point.
- **SHA (Sidereal Hour Angle):** Offset from Aries GHA to the star's GHA, measured westwards.

Given these definitions, we can calculate the longitude of a star GP by:

$$\lambda = 360^\circ - (\text{Aries GHA} + \text{SHA})$$

If the longitude goes over  $180^\circ$  or under  $-180^\circ$ , we can simply subtract or add  $360^\circ$  to keep it within  $[-180^\circ, 180^\circ]$ . However, this adjustment won't affect the results, so feel free to skip it.

Latitude is simply the star's declination.

## Example Star GPs

### Regulus:

- SHA:  $207^\circ 39.6'$
- DEC:  $11^\circ 52.5'$
- Aries GHA:  $181^\circ 26.6'$
- GHA:  $29^\circ 06.2'$
- GP:  $(11^\circ 52.5', -29^\circ 06.2')$

### Arcturus:

- SHA:  $145^\circ 52.4'$
- DEC:  $19^\circ 05.2'$
- Aries GHA:  $182^\circ 00.4'$
- GHA:  $327^\circ 52.8'$
- GP:  $(19^\circ 05.2', 32^\circ 07.2')$

### Dubhe:

- SHA:  $193^\circ 47.3'$
- DEC:  $61^\circ 39'$
- Aries GHA:  $182^\circ 26.7'$
- GHA:  $16^\circ 14'$
- GP:  $(61^\circ 39', -16^\circ 14')$

## Circles of Equal Altitude

Circles of equal altitude are circles where, everywhere along the circumference, the star would be observed at the same altitude angle.

To define a circle of equal altitude, we first determine its northernmost point by adding the zenith angle ( $90^\circ - \text{alt}$ ) to the GP latitude, and its southernmost point by subtracting it. The longitude remains the same as the GP. Then we convert these to Cartesian coordinates.

For Regulus we obtain:

- Northernmost point:  $(0.493703, -0.274829)$
- Southernmost point:  $(0.993046, -0.552798)$

If latitude exceeds  $90^\circ$ , it means the northern point is on the other side of the pole. No adjustment is needed, since the distance formula naturally produces a negative value in this case.

The circle center is the midpoint of these two coordinates:

$$(0.743375, -0.413814)$$

The radius is half the distance between the two points:

$$r = 0.285749$$

Repeating this process for the other stars:

**Arcturus:**

- Center: (1.021775, 0.641455)
- Radius: 1.135954

**Dubhe:**

- Center: (0.267833, -0.077982)
- Radius: 0.334367

## Circle Intersections

Circle intersections can be solved with several methods (or by using Geogebra). Here is a step-by-step algebraic method:

1. Let the circle centers be  $A(a_x, a_y)$  and  $B(b_x, b_y)$ , with radii  $ar$  and  $br$ .
2. Compute the distance between the centers using the Pythagorean theorem:

$$d = \sqrt{(b_x - a_x)^2 + (b_y - a_y)^2}$$

3. Consider the line connecting the two intersection points. Let  $P$  be the point where this line crosses the line segment  $AB$ . The distance from  $A$  to  $P$  is:

$$t = \frac{ar^2 + d^2 - br^2}{2d}$$

4. The distance from  $P$  to either intersection point is:

$$h = \sqrt{ar^2 - t^2}$$

5. Compute the direction cosines of the line from  $A$  to  $B$ :

$$\cos \theta = \frac{b_x - a_x}{d}, \quad \sin \theta = \frac{b_y - a_y}{d}$$

6. Finally, the coordinates of the two intersection points are:

$$(x, y)' = (a_x + t \cos \theta - h \sin \theta, a_y + t \sin \theta + h \cos \theta)$$

$$(x, y)'' = (a_x + t \cos \theta + h \sin \theta, a_y + t \sin \theta - h \cos \theta)$$

For the first two circles we find:

- (0.465128, -0.348765)
- (1.017498, -0.494491)

## Inverse Conversion

To convert Cartesian  $(x, y)$  back to longitude and latitude:

**Longitude:**

$$\lambda(x, y) = \arctan 2(y, x)$$

**Latitude:**

First compute distance from the pole:

$$d(x, y) = \sqrt{x^2 + y^2}$$

Then apply:

$$\varphi(d) = 2 \times \arctan\left(\frac{1 - y}{1 + y}\right)$$

This formula is simply the inverse of the one that converts latitude into distance from the pole.

## Results

Applying these formulas to the intersections yields the coordinates:

- (29.655859, -36.863503)
- (-7.050214, -25.91914)

When combining all pairs of circles, we obtain six intersection points. However, not all of them are consistent: three of the points lie close together, while the others are far away. The cluster of three nearby points represents the true observer's location, since the correct solution should appear repeatedly across different circle pairs.

These three consistent points are:

- (29.655859, -36.863503)

- $(29.718368, -36.872985)$
- $(29.691066, -36.770559)$

To refine the estimate and reduce small errors from measurement, we take the average of these three coordinates. This gives the final estimated position:

$(29.688431, -36.835682)$

The values shown here are rounded to six decimals for readability. However, in the calculations I used the full precision available on the calculator. If one repeatedly rounds values while replicating the method, the results may diverge due to accumulated rounding errors.